For each media category and lexical measure, the results were binned into years and the median taken each year. The median was used to reduce the effect of outliers (similar results were found when using the mean). These were plotted on a scatterplot (see Supplementary Information).

Visually, the scatterplots are suggestive of some change in the gradient of the lexical measure in time. In order to estimate the location of these breakpoints, we used python's piecewise-regression package [31] with default settings. The regression fits and locations of breakpoints are shown in the scatterplots in the Supplementary Information.

We ran a similar analysis with the categories combined. In order to combine the categories, we first took means for each year and category and then took the mean across categories for each year. It is more natural to use means than medians when combining categories, and the influence of outliers is smaller as there is more data than in the individual categories. The scatterplot and piecewise-regression fit for the combined word entropy is shown in the Supplementary Information.

Timeseries Trend Analysis

For each category and lexical measure, trend analyses were carried out on the annual median values. This was done between the years 1900 and 2009 (the last year of data). KPSS and MK tests were carried out for each measure and media category in COHA (full results in Supplementary Information).

The Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test assumes the null hypothesis of a stationary timeseries. p-values below 0.05 mean that we can reject this hypothesis at 5% significance and provide evidence of a trend. The test was applied using python's statsmodels package [32].

The Mann-Kendall (MK) test is a non-parametric trend test [33]. The test assumes no serial correlation i.e. errors in one observation do not predict errors in other observations [33]. The text corpora are constructed from independent text samples so this is a reasonable assumption. The null hypothesis is that the data has no trend, and the p-value tells us the probabilty that the data was observed under the null hypothesis. At 5% significance we reject the null hypothesis if $p < 0.05$. The test was carried out using python's pymannkendall package [33]

In the Supplementary Information we calcaulte Pearson's R between magazine circulation and word entropy.

Timeseries Smoothing

While we included scatterplots for annual binned data in the Supplementary Information, the trends are easier to see visually with a smoothed timeseries. For Figures 2 and 3 the timeseries was smoothed using a moving average with measures of text samples from $\pm$ 5 years. The 95% confidence interval was calculated as the standard error of this mean calculation multiplied by 1.96 (assuming normally distributed errors). For each lexical measure, the mean was plotted for each year with the confidence interval region shaded. We only included years where we had a minimum of 10 data points within the window.

We report the smoothed timeseries for each of the COHA text categories, as well as the categories combined. The timeseries for media categories were combined by taking an average across the timeseries annual means for the media categories that had a value for that year. The 95% confidence interval was again calculated as 1.96 times the standard error. For each year, the standard error of the estimate of the mean, $SE_{\bar{X}}$ was computed based on the delta method,

$$SE_{\bar{X}} = \frac{\sqrt{\sum_{i=1}^{n} SE_i^2}}{n}\,, \tag{4}$$

with n depending on how many media categories had values for the annual mean each year.

Differences Between Media Categories

We looked at the distributions of the lexical measures within media categories in COCA, the BNC and COHA (restricted to 2000-2007 to avoid the effect of historical changes). To test for differences between the groups we carried out ANOVA tests across categories within each corpora separately for each of the lexical measures. At 5% significance, $p < 0.05$ provides evidence that the media categories are drawn from different underlying population distributions. The tests were carried out using python's statsmodels package [32]

For visualisation, the distributions of word entropy for each media category are shown as a kernel density estimate with the bandwidth determined by the Scott rule and the density trimmed to the data range.

US Magazine Circulation

The data for magazine circulation numbers (reported in the Supplementary Information) were taken from Sumner's "The Magazine Century American Magazines Since 1900" [34] Chapter 1, which are

attributed to data originally from the Audit Bureau of Circulation. This data source does not track all US magazines, but does track well-known magazines. The data was plotted without further treatment.

Results

The Rising Entropy of American English

We analysed the Corpus of Historical American English (COHA), a balanced corpus with text samples from the 1810s to the 2000s categorised into news, magazines, fiction and non-fiction [21]. As discussed in the Methods section, we analysed text samples truncated to $N = 2000$ words. We found a clear trend of rising lexical diversity since approximately 1900 as measured by word entropy, type token ratio and Zipf exponent (Figure 1).

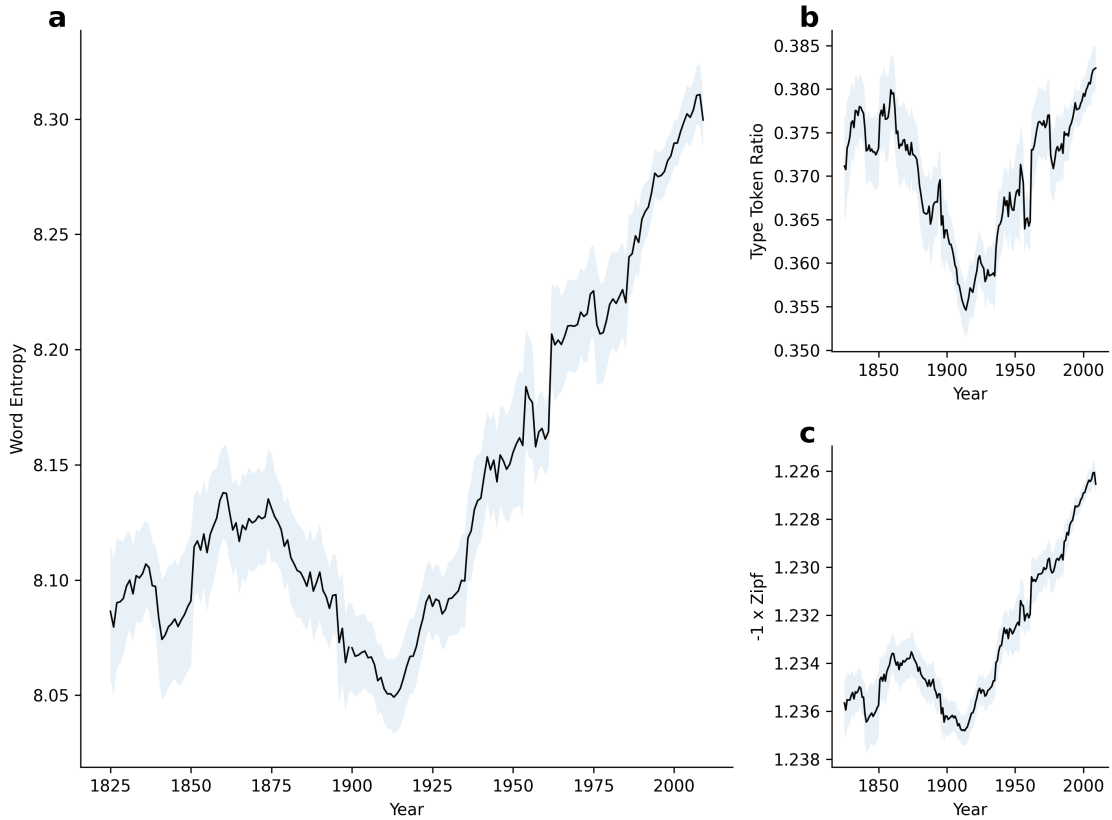

Figure 1: Lexical diversity of text samples in the Corpus of Historical American English as measured by a) word entropy, b) type token ratio and c) Zipf exponent. Timeseries are smoothed with a moving average window of $\pm$ 5 years, and averaged over media categories. Shaded region shows 95% confidence interval of this average.

The trends in separate media categories follow the same pattern of rising lexical diversity as measured by word entropy (Figure 2). We analysed the timeseries of annual averages since 1900 for each media category (fiction, non-fiction, news, magazines) and lexical measure (word entropy, Zipf exponent, type token ratio) using Kwiatkowski–Phillips–Schmidt–Shin (KPSS) and Mann-Kendall (MK) tests on the annual median values (using the annual mean gives similar results). This gives a total of $4 \times 3 \times 2 = 24$ trend tests. All 24 tests show significant evidence of a trend at $p < 0.05$ And 22 out of 24 tests show significant evidence of trends at $p < 0.01$ (the tests for a trend in type token ratio in non-fiction had KPSS $p = 0.015$ and MK $p = 0.013$). Overall there is very strong evidence for a trend of rising lexical diversity in all media categories between 1900 and 2010. For full results and a deeper analysis, see the Supplementary Information.

Figure 2: Timeseries of word entropy across media categories in the Corpus of Historical American English. For each media category, the timeseries was smoothed using an average over a window of $\pm$ 5 years. The shaded regions are 95% confidence intervals of this average. All media categories show an upward trend in word entropy from 1900.

Higher Entropy in Short-form Media

The historical trend (Figure 2) suggests modern differences in entropy between media categories. However, we also know that short-form media has become especially prominent with the recent rise of online platforms for media distribution, such as social media, RSS feeds, and news platforms that present short headlines and snippets that link to long-form articles. To investigate these different media categories, we examined the Corpus of Contemporary American English (COCA) and the British National Corpus (BNC), as well as social media data from Twitter and Reddit. Figure 3 shows the distribution of word entropy across different media categories. Within COHA (limited to 2000-2007), BNC, and COCA there were significant differences in all lexical measures across media categories (ANOVA tests $p < 0.01$). Full statistical results are in the Supplementary Information. Overall, short-form media categories of news and magazines have higher entropy than long-form media, and social media feeds have the highest entropy of all.

It should be noted that when analysing social media data we collated posts to create text samples with $N = 2000$ words, to match the length of the other media type analyses. Combining posts will naturally lead to high entropy text, with fast switching of contexts and high novelty. This mirrors how people actually consume social media. Essentially, social media platforms generate high entropy information environments in the form of feeds of short messages from different users. This is not necessarily a linguistic change in how people generate English; it is a change in how people consume English text.

Information Foraging in the Attention Economy

The results are suggestive of a link between competition for attention and word entropy. To explain these results we generate a model of the attention economy based on information foraging. Foraging